

$$\textcircled{a} \quad f'(-1) = \lim_{h \rightarrow 0} \frac{f(-1+h) - f(-1)}{h} = \lim_{h \rightarrow 0} \frac{(-1+h)^3 - (-1)^3}{h} = 3$$

1. Consider the function $f(x) = x^3$; $x_0 = -1$ and $x_1 = 0$.

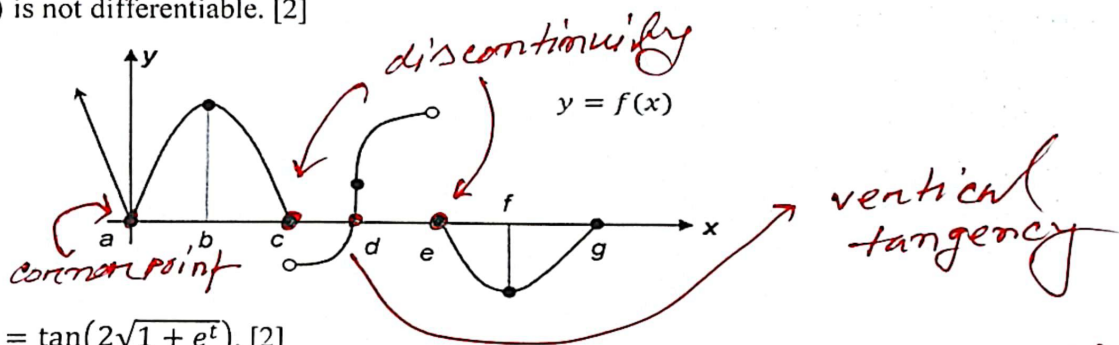
(a). Using definition find the instantaneous rate change of function at x_0 . [2]

(b). Find the average rate of change of function in the interval $[x_0, x_1]$. [1]

(c). Find the equation of the secant line of $f(x)$ within the specified interval. [2]

$$\textcircled{b} \quad y - f(-1) = 1(x - (-1)) \Rightarrow y - (-1) = x - (-1) \Rightarrow y = x = 1$$

2. The graph of function $f(x)$ is given below. State with explanation the locations (value of x) at which $f(x)$ is not differentiable. [2]



3. Find $\frac{dw}{dt}$, if $w = \tan(2\sqrt{1+e^t})$. [2]

4. Find $\frac{dw}{dt}$, where $w = x^2$, $y = \sin^{-1} x$, $x = \ln t$. [2] (Wrong question, $w = y^2$)

$$\textcircled{3} \quad \frac{dw}{dt} = \sec^2(2\sqrt{1+e^t}) \left(2 \cdot \frac{1}{2\sqrt{1+e^t}} \right) (0 + e^t)$$

$$\textcircled{4} \quad \frac{dw}{dt} = \frac{dw}{dx} \frac{dx}{dt} = (2x) \left(\frac{1}{t} \right) = \frac{2 \ln t}{t}$$

$$\frac{dw}{dt} = \frac{dw}{dy} \frac{dy}{dx} \frac{dx}{dt} = 2y \cdot \frac{1}{\sqrt{1-x^2}} \cdot \frac{1}{t} = \frac{2 \sin^{-1}(\ln t)}{t \sqrt{1-(\ln t)^2}}$$

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